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15 January 1980
SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, KNITTED, POLYESTER, DOUBLE KNIT (LA COSTE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for a La Coste double knit polyester cloth of one type only.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

(Application for copies of the AATCC Manual should be addressed to the AATCC National Headquarters, PO Box 12215, Research Triangle Park, NC 27709.)

3. REQUIREMENTS.

3.1 Standard sample. The finished cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials. (See 6.4)

3.3.1 Fiber. The fiber shall be polyester glycol terephthalate, regular tenacity, semi-dull. (see 4.4.1).

*3.3.2 Yarn. The yarn shall be nominal 150 denier polyester, 33 ± 3 filaments, and shall be texturized by the false twist process (see 4.4.1).

* 3.4 Physical requirements. The finished cloth shall conform to the physical requirements specified in Table I, when tested as specified in 4.5.

Table I - Physical requirements

Weight oz/sq yd (min)	Weight oz/sq yd (max)	Weight (g/m^2) (min)	Weight (g/m^2) (max)	Wales/inch (cm) (min) 1/	Courses/inch (cm) (min) 1/	Bursting strength lbs. (min)	Bursting strength Newton (min)
6.5	7.0	(220.4)	(237.3)	30 (12.0)	28 (11.0)	200	(890)

1/ Count wales and courses on back of cloth.

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* 3.4.1 Knit. The cloth shall be of a double knit La Coste construction (to match the standard samples), and knitted on a minimum 22 cut circular knitting machine (see 4.4.1) The cloth shall be split (open width) prior to finishing.

* 3.4.2 Knit pattern. The cloth shall be knitted on a double knit machine as specified in 3.4.1 with multiples of six feeds with the following pattern: (See 4.4.1)

Feed	Dial	Cylinder
1	Tuck (Short needles)	Knit (all)
2	Knit (Short needles)	Welt
3	Knit (Short needles)	Welt
4	Tuck (Long needles)	Knit (all)
5	Knit (Long needles)	Welt
6	Knit (Long needles)	Welt

3.4.3 Width. Unless otherwise specified (see 6.2), the split width of the finished cloth shall be a minimum of 62 inches (157.5cm).

3.5 Color. Unless otherwise specified (see 6.2), the color of the finished cloth shall be White 3006 and shall match the standard sample. The cloth shall be fully bleached and supplemented with a blue violet fluorescein optical brightener. All cloth within any one contract shall be processed for whiteness in the same manner.

3.5.1 Matching. The finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500 Kelvin and shall be a good approximation to standard sample under incandescent lamplight at 2800 Kelvin.

3.5.2 Colorfastness. The finished cloth shall show fastness to perspiration, laundering and light equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available the cloth shall show a minimum of "good" fastness to perspiration, laundering and light when tested as specified in 4.5.

3.6 Finish. The cloth shall be scoured, and heat set. (See 4.4.1) The character of the finish shall be equal to that of the standard sample. The face of the cloth shall have the honey comb effect.

3.6.1 Nonfibrous material. The starch and protein content, including chloroform-soluble and water-soluble material, shall not exceed 2.0 percent when tested as specified in 4.5.

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3.7 Dimensional stability. The finished cloth shall not shrink or elongate more than 3.0% in the wale or more than 2.0% in the course direction, when tested as specified in 4.5.

3.8 pH. The pH value of the water extract of the finished cloth shall be not lower than 5.0 nor higher than 8.5 when tested as specified in 4.5.

3.9 Face marking. The face side of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous pieces each not less than 50 yards (45.7m) in length and shall be put up in rolls, full width, face inside, as specified in PPP-P-1133.

3.11 Fiber identification. Each roll of cloth shall be labeled, or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (See 4.3).
2. Quality conformance inspection (See 4.4).

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4.3 First article inspection. When required (see 6.2), the first article of finished cloth shall be visually examined for color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the contractor shall furnish a certificate of compliance for the fiber requirements stated in 3.3.1, yarn requirement of 3.3.2, knitting machine requirements stated in 3.4.1, knit pattern stated in 3.4.2, and finish stated in 3.6.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

* 4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined on the face side and visual defects as defined in MIL-STD-1491, shall be classified as listed. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (lengthwise) yard (0.91m) or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard (0.91m). The acceptable quality levels for this examination shall be 2.5 major defects and 6.5 total defects (major and minor combined) per 100 units (yards)(0.91m). The sample size shall be based upon inspection level II of MIL-STD-105. The lot size shall be expressed in units of one yard (0.91m). The number of rolls from which the sample is to be selected shall be in accordance with Table II. An approximately equal number of yards shall be examined from each roll in the sample.

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*Defects	Classification Major	Minor
Any hole, cut, tear, run	X	
Dropped stitch, end out	X	
Tuck stitch, pulled stitch or needle streak		X
Mend (any)	X	
Coarse or fine yarns 1/	X	
Foreign matter 1/		X
Knots oversize, extending above the surface of cloth	X	
Kinks or untrimmed ends		X
Crease or wrinkle, embedded		X
Stretch (stitch tension) knitting tension too tight or too loose	X	
Any spot or stain	X	
Shade bars	X	
Barre marks	X	
Any break in knit pattern	X	

1/ Clearly visible at normal inspection distance (approximately 3 feet)(0.91m)

4.4.2.2 Overall examination. Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be as shown in Table II.

*Defects

Face side of cloth not on inside.
Width less than specified.
Overall uncleanness.
Uneven knitting throughout the piece.
Cut edge is not parallel to wales in finished cloth.
Character of finish not equal to standard.

Table II - Sample sizes

Lot size (Yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

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4.4.2.3 Examination for length.

4.4.2.3.1 Individual rolls. The rolls shall be examined for gross length. Any gross length found to be less than the minimum specified or more than 2 yards (1.82m) less than the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table II.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.4 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.5 Examination for face identification, identification of preshrinkage process and non-compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with the section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 2-1/2 yards (2.3m) full width of the finished cloth. The sample size shall be as listed in Table III. The lot size shall be expressed in units of 1 yard (0.91m). The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

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Table III - Sample size

Lot size (yards)(0.91m)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table IV - Test methods

Characteristic	Requirement paragraph	Test method
Weight	3.4	5041
Strength (Bursting)	3.4	5120
Wales and Courses	3.4	5070
Colorfastness to:		
Perspiration	3.5.2	5680
Laundering	3.5.2	5610
Light	3.5.2	5660
*Nonfibrous material	3.6.1	2611
Dimensional stability	3.7	4.5.1
pH	3.8	2811

4.5.1 Shrinkage to laundering. The finished cloth shall be tested after 5 cycles according to AATCC Test Method 135-1973 Method II (120 ± 5°F - 49 + 3°C), Drying Procedure "B" (Tumble).

5. PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C. The cloth put-up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirement of PPP-P-1133.

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES.

6.1 Intended use. The cloth covered by this specification is intended for use as the basic material for the pant suit for female nurses.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth, if other than specified (see 3.5).
(c) When first article is required (see 3.2 and 4.3).
(d) Width of cloth required (see 3.4.1) if other than specified.
(e) Length, if other than specified (see 3.7).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requrements of this specification.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-87044A Cloth, Knitted, Polyester, Double Knit (La Coste)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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